The Looking-Glass Transform: An Exact Equivalence Between the Schrödinger and Liouville Equations

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Abstract

The Schrödinger equation is exactly equivalent to the classical Liouville equation in a modified potential. Lifting classical Hamiltonian dynamics to a conservation law for the phase-space distribution, Fourier transforming in momentum, and replacing the potential by its *looking-glass potential*—its average over the segment between the two points $x \pm \hat{p}/2$, where $\hat{p}$ is the variable conjugate to momentum—yields, with no remainder, an equation that separates into coupled Schrödinger equations; the phase-space distribution is the Wigner quasi-probability function, and the Born position density is its momentum marginal. The ingredients of this route are classical results of the phase-space formulation of quantum mechanics, and closely related Liouville-to-Schrödinger constructions exist. The contribution here is the framing and its consequences: the segment average is a finite centered difference of the potential, kept finite rather than taken as a limit, so quantum dynamics is classical dynamics with the force averaged over the coherence length of the state. The looking-glass potential differs from the bare potential by a series whose leading term, $(\hbar^2/24)\,\partial^3 V/\partial x^3\,\partial^3 w/\partial p^3$ in phase space, is the Moyal correction read in reverse; the series vanishes in effect for quadratic potentials and terminates after that term for quartic ones. Two observations follow from this viewpoint: Wigner negativity admits a classical reading as fluctuation below a uniform background, since $w \to w + c$ is invisible to the Liouville flux; and the transformed dynamics is well posed on a state space strictly larger than the quantum one, which contains the classical point preparations excluded by the uncertainty principle. For a separated entangled pair measured by an apparatus that interacts only with the distant particle, every Hamiltonian is quadratic, so classical flow in the bare potential is already exact, and collapse is conditioning of the signed joint distribution on the pointer marginal: the flow is local, the Born weights emerge as the apparatus grows, and—unlike classical conditioning—conditioning can create negativity. Classical flow

in the bare potential is a good approximation exactly where the two potentials are close; where they differ appreciably—anharmonic potentials at long times, and states with long coherence lengths such as energy eigenstates—experiment follows flow in the looking-glass potential, as the equivalence requires. The value of the transform is structural and pedagogical: it locates exactly what must be added to probabilistic classical mechanics to obtain quantum mechanics—one replacement of the potential by its segment average—in the language of Fourier transforms and finite differences.

1 Introduction

Quantum mechanics works extraordinarily well, yet its interpretation remains contested. Why does the wavefunction give only probabilities? What happens during measurement? Does the universe branch into many worlds to account for each outcome [1]? These questions persist in part because the standard formulation takes the Schrödinger equation and the Born rule as postulates.

The phase-space formulation of quantum mechanics [2, 3, 4, 5] makes the kinship with classical mechanics most visible: the quantum evolution of the Wigner function differs from the classical Liouville flow only by a series of terms beginning at order $\hbar^2$, and is identical to it for quadratic potentials [6]. That correspondence is usually read from quantum to classical: start with a wavefunction, construct the Wigner function, and recover classical mechanics as $\hbar \to 0$.

This paper reads it in the other direction. We start from classical Hamiltonian mechanics with probabilistic initial data and apply three elementary steps: lift the equations of motion to a conservation law for the phase-space distribution, Fourier transform in momentum, and replace the potential gradient by its average over a finite segment. At leading order that average is a centered difference of the potential; in full it is the gradient of the *looking-glass potential*, the average of V over the segment, and with it the result separates exactly into coupled Schrödinger equations. We call the composite the looking-glass transform.

The reverse direction has been explored before. Carnovali and França [7] derive a Schrödinger-type equation from the Liouville equation by an equivalent route: they replace the force at the midpoint of the segment by its average over the segment, which integrates to the difference of the potential at the end points (their mean-value step), and justify the replacement in the limit in which the end points coincide. Bondar et al. [8] obtain both quantum and classical (Koopman–von Neumann [9]) dynamics from operational axioms, with the Wigner function itself serving as the wave function. Hall, Deckert, and Wiseman [10] build quantum behavior from many interacting classical worlds by entirely different machinery.

The present treatment keeps the segment finite and keeps the remainder exactly: quantum dynamics is classical Liouville dynamics plus one explicit correction term (Section 6.1). Its leading-order form is the centered difference of V, which is exact for quadratic potentials and whose first correction is the Moyal $\hbar^2$ term; the segment average generalizes the centered difference to all orders, so nothing is removed by a limit.

We are explicit about what is and is not new here. The transform itself is not: the route from the Liouville equation through a momentum Fourier transform and shifted variables to coupled Schrödinger equations was published by Carnovali and França [7], and the underlying correspondence is the classical Wigner–Moyal one [6, 5]. What we offer is a particular reading of that route and four observations that follow from it. The first is organizational rather than novel: the entire difference between the dynamics of probabilistic classical mechanics and quantum mechanics is one explicit replacement—the potential by its average over the segment between the two branch points—whose leading-order form is the centered difference and whose expansion is the Moyal series read in reverse. The segment is kept finite rather than shrunk by the limit taken in [7], which turns the replacement from an approximation into an exact equivalence, and its length—the coherence length of the state—sets how far quantum dynamics departs from classical flow in the bare potential. The second is a reading of Wigner negativity: a uniform background $w \to w + c$ is invisible to the Liouville flux, so w behaves like an acoustic pressure perturbation, and its negative values are rarefactions below an ambient level rather than negative probabilities. Negativity is usually treated as the irreducibly quantum feature of w [11]; Section 8.4 argues that for a model whose predictions—its marginals—are non-negative and correct, it does not matter. The third is structural: the transformed dynamics is well posed on a state space larger than the quantum one—large enough to contain the classical point preparations the uncertainty principle excludes—within which the quantum states, those with positive-semidefinite C, form an invariant subset. The constituent facts here are known (the bounds on valid Wigner distributions are classical [12]); what we add is only the reading of the larger set as a dynamical arena. The fourth concerns the collapse of a separated entangled pair (Section 8.3). For free particles and a von Neumann pointer coupled only to the distant particle, every Hamiltonian is quadratic, so the looking-glass potential exerts the same force as the bare one, and measurement is classical Liouville flow of a signed distribution followed by conditioning on the pointer marginal. The flow is local, nothing done at the distant particle signals, and the Born weights emerge as the apparatus grows; the quantum ingredient is confined to the sign of the initial distribution, and conditioning with signed weights can create negativity that classical conditioning never could. The underlying identity—that conditioning the joint Wigner function reproduces the post-measurement state—is standard, and Bell made the classical-conditioning point

for positive Wigner functions [13]; what we add is the exact, fully classical account of the measurement interaction and its numerical illustration.

Two facts frame the view: the phase-space distribution $w(x, p, t)$ is the Wigner function [2]—a signed field satisfying a local conservation law whose marginals are genuine probabilities—and where the looking-glass and bare potentials differ appreciably, experiment follows Liouville flow in the looking-glass potential—the quantum dynamics—not flow in the bare one (Section 6). Within this representation the Born position density is not an additional postulate but the momentum marginal of w; what this does and does not explain about measurement is taken up in Section 7.

We work in one spatial dimension for clarity; the extension to higher dimensions and multiple particles is straightforward.

2 Quantum Perspective

Consider a tiny one-dimensional non-relativistic particle of mass M in a tiny potential $V(x, t)$. Here the accepted model is the Schrödinger equation, where $|\Psi(x, t)|^2$ gives the probability of observing the particle at location x at time t. The wave function $\Psi(x, t)$ evolves according to the partial differential equation [14]

$$i\hbar\Psi_t = -\frac{\hbar^2}{2M}\Psi_{xx} + V\Psi\,, \tag{1}$$

where $\hbar \approx 1.05 \times 10^{-34}$ joule-seconds.

3 Classical Perspective

Consider a regular one-dimensional non-relativistic particle of mass M in a regular potential $V(x)$. The position of the particle $x(t)$ and momentum of the particle $p(t)$ evolve according to Hamilton's equations [15]

$$M\frac{dx}{dt} = p\,, \tag{2}$$

$$\frac{dp}{dt} = -\frac{\partial V}{\partial x}\,. \tag{3}$$

4 Three Concepts to Get Quantum Mechanics from Classical Mechanics

What must be added to classical mechanics to get quantum mechanics? We combine the following concepts.

4.1 An ordinary differential equation can be lifted to a linear partial differential equation

Think of an ordinary differential equation initial value problem as evolving a partial differential equation which describes the evolution of the probability distribution of the initial values [16].

ODE:

$$\frac{du_1}{dt} = v_1\,, \tag{4}$$

$$\frac{du_2}{dt} = v_2\,, \tag{5}$$

$$\vdots \tag{6}$$

$$\frac{du_n}{dt} = v_n\,. \tag{7}$$

PDE:

$$\frac{\partial w}{\partial t} + \frac{\partial (v_1 w)}{\partial u_1} + \frac{\partial (v_2 w)}{\partial u_2} + \cdots + \frac{\partial (v_n w)}{\partial u_n} = 0\,. \tag{8}$$

Using the Dirac delta function for the initial value, the PDE recovers the ODE initial value problem: $w(u,0) = w_0(u) = \delta(u - u_0)$.

4.2 The Fourier transform

$$\hat{\Psi}(\hat{p}) = \int_{-\infty}^{+\infty} dp\, e^{-i\frac{\hat{p}p}{\hbar}}\, \Psi(p)\,, \tag{9}$$

$$\Psi(p) = \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar}\, e^{+i\frac{\hat{p}p}{\hbar}}\, \hat{\Psi}(\hat{p})\,, \tag{10}$$

$$\partial_p \Psi \to i\frac{\hat{p}}{\hbar}\hat{\Psi}\,, \tag{11}$$

$$p\Psi \to i\hbar\partial_{\hat{p}}\hat{\Psi}\,. \tag{12}$$

$$\delta(\hat{p}) = \int_{-\infty}^{+\infty} \frac{dp}{2\pi\hbar}\, e^{i\frac{\hat{p}p}{\hbar}}\,. \tag{13}$$

The use of $\hat{p} = \hbar\omega$ compared to the typical spectral parameter ω greatly compresses the spectrum.

4.3 Centered-difference approximation of $\frac{\partial V}{\partial x}$

From the Taylor expansion of V around x, we have

$$\frac{\partial V}{\partial x} \approx \frac{V(x+\varepsilon/2,t) - V(x-\varepsilon/2,t)}{\varepsilon} + O(\varepsilon^2)\,. \tag{14}$$

We actually approximate $\varepsilon \cdot \frac{\partial V}{\partial x}$, so with first error term:

$$\varepsilon\frac{\partial V}{\partial x} \approx V(x+\varepsilon/2,t) - V(x-\varepsilon/2,t) - \frac{\varepsilon^3}{24}\frac{\partial^3 V(x,t)}{\partial x^3} + O(\varepsilon^5)\,. \tag{15}$$

5 Quantum from Probabilistic Classical Mechanics: The Looking-Glass Transform

We now compose the three concepts. The result reveals the Schrödinger structure *inside* classical mechanics. We present the leading-order form, a centered difference of the potential, whose error term is given in Section 6; Section 6.1 gives the exact remainder in closed form.

We start with the classical Hamiltonian ordinary differential equation:

$$\frac{dx}{dt} = \frac{1}{M}p\,, \tag{16}$$

$$\frac{dp}{dt} = -\frac{\partial V}{\partial x}\,. \tag{17}$$

Using the conservation of probability, this corresponds to the following partial differential equation describing the evolution of the probability density:

$$\frac{\partial w}{\partial t} + \frac{\partial}{\partial x}\left(\frac{1}{M}pw\right) + \frac{\partial}{\partial p}\left(-\frac{\partial V}{\partial x}w\right) = 0\,, \tag{18}$$

or

$$w_t + \frac{p}{M}w_x - \frac{\partial V}{\partial x}w_p = 0\,. \tag{19}$$

Now Fourier transform only with respect to the momentum p and multiply by $i\hbar$. This results in

$$i\hbar\hat{w}_t - \frac{\hbar^2}{M}\hat{w}_{x\hat{p}} + \hat{p}\frac{\partial V}{\partial x}\hat{w} = 0\,. \tag{20}$$

We now change variables. Define

$$z_\pm = x \pm \frac{\hat{p}}{2}\,, \tag{21}$$

So

$$\frac{\partial}{\partial x} \;\to\; \frac{\partial}{\partial z_+} + \frac{\partial}{\partial z_-}\,, \tag{22}$$

$$\frac{\partial}{\partial \hat{p}} \;\to\; \frac{1}{2}\frac{\partial}{\partial z_+} - \frac{1}{2}\frac{\partial}{\partial z_-} \tag{23}$$

This transforms (20) into

$$i\hbar\hat{w}_t - \frac{\hbar^2}{2M}\left[\left(\frac{\partial}{\partial z_+}\right)^2 - \left(\frac{\partial}{\partial z_-}\right)^2\right]\hat{w} + \hat{p}\frac{\partial V}{\partial x}\hat{w} = 0 \tag{24}$$

Next is the only approximation. (Section 6.1 gives the exact remainder.) We use (14) with $\varepsilon = \hat{p}$:

$$\hat{p}\frac{\partial V}{\partial x} \approx V(x+\hat{p}/2,t) - V(x-\hat{p}/2,t) \tag{25}$$

This transforms (20) into

$$i\hbar\hat{w}_t - \frac{\hbar^2}{2M}\left[\left(\frac{\partial}{\partial z_+}\right)^2 - \left(\frac{\partial}{\partial z_-}\right)^2\right]\hat{w} + [V(z_+,t) - V(z_-,t)]\hat{w} = 0\,. \tag{26}$$

While this does not yet look like Schrödinger's equation, this is a linear differential equation, and we can look at separation of variables to construct a general solution to (26). Suppose $\hat{w} = \sum \psi^+(z_+,t)\psi^-(z_-,t)$ where each of these terms satisfy the above, we have

$$\mp i\hbar\psi^\pm_t = -\frac{\hbar^2}{2M}\psi^\pm_{zz} + V(z_\pm,t)\psi^\pm\,. \tag{27}$$

Here z takes the respective roles of z_+ and z_-.

For real potential $V(x,t)$, $\psi^+(z_+,t) = \Psi^*(z_+,t)$ and $\psi^-(z_-,t) = \Psi(z_-,t)$, where $\Psi(x,t)$ is the original Schrödinger wavefunction (1).

If additionally $V(x,t)$ is time-independent, then $\psi^+ = e^{+i\lambda t}\psi^*_\lambda$ and $\psi^- = e^{-i\lambda t}\psi_\lambda$ where the ψ_λ are eigenvectors of the original Schrödinger's equation (1). These form a complete basis for the solution of PDE for $\hat{w}$:

$$\hat{w}(t,z_+,z_-) = \sum_{\lambda,\lambda'} C_{\lambda,\lambda'} e^{i(\lambda-\lambda')t}\psi^*_\lambda(z_+)\psi_{\lambda'}(z_-)\,. \tag{28}$$

where the $C_{\lambda,\lambda'}$ are determined by the initial distribution w_0. Assuming ψ_λ are normalized by $\int_{-\infty}^{+\infty} dz\,|\psi_\lambda(z)|^2 = 1$, we have:

$$C_{\lambda,\lambda'} = \int_{-\infty}^{+\infty} dz_+ \int_{-\infty}^{+\infty} dz_-\,\psi_\lambda(z_+)\psi^*_{\lambda'}(z_-)\hat{w}_0(z_+,z_-)\,, \tag{29}$$

$$= \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} d\hat{p}\,\psi_\lambda(x+\frac{\hat{p}}{2})\psi^*_{\lambda'}(x-\frac{\hat{p}}{2})\hat{w}_0(x,\hat{p})\,. \tag{30}$$

Note C is Hermitian and w is normalized. This implies $C = \sum_k c_k a_k^\dagger a_k$ for real eigenvalues c_k, unit left eigenvectors a_k (so $a_k C = c_k a_k$), and $\sum_k c_k = 1$.

The quantity $w(x,p,t)$ is precisely the Wigner quasi-probability distribution [2], and $\hat{w}$ is its Fourier transform in p. In the quantum-to-classical direction, the exact evolution generated by $V(z_+,t) - V(z_-,t)$ and its expansion into the classical force term plus $O(\hbar^2)$ corrections are standard results [6, 5]. The looking-glass transform runs that arrow in reverse: rather than starting from a wavefunction and constructing w, we started from the classical Liouville equation and found that a single centered difference converts its dynamics into Schrödinger dynamics. The mathematics of the correspondence is the classical Wigner–Moyal one; the contribution of this paper is the direction in which it is read, and what that reading buys.

Intuition. Quantum mechanics viewed through the lens of the Schrödinger equation carries essentially no classical intuition. Here is a partial dictionary between the two worlds. Any classical distribution evolves, to $O(\hbar^2)$, like a quantum-mechanical structure—though not every classical distribution is a quantum state; Section 8.1 returns to this. The reverse direction takes one additional mental leap: a uniform background probability. $w \to w + c$ for some constant c solves the local probability flux equation (19) just fine. This is the nod to quasi-probability. For many interesting Quantum configurations, the corresponding real quasi-probability,

$$w(x,p,t) = \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar}\, e^{+i\frac{\hat{p}p}{\hbar}}\,\Psi^*(x+\frac{\hat{p}}{2},t)\Psi(x-\frac{\hat{p}}{2},t) \tag{31}$$

is real-valued, integrates to 1, but not necessarily non-negative everywhere. Read it like a pressure perturbation about a uniform ambient level c: negative values are rarefactions, not missing probability, and classical intuition carries over (Section 8.4).

6 Error Estimate

The centered-difference approximation (14) is exact for linear and quadratic potentials. For other potentials, the leading error term is

$$\hat{p}\frac{\partial V}{\partial x} \approx V(x+\frac{\hat{p}}{2}) - V(x-\frac{\hat{p}}{2}) + \left[-\frac{\hat{p}^3}{24}\frac{\partial^3 V(x)}{\partial x^3}\right] . \tag{32}$$

For the approximation to be closer to exact, then the PDE (19) would have to be modified to

$$w_t + \frac{p}{M}w_x - \frac{\partial V}{\partial x}w_p = \left[-\frac{\hbar^2}{24}\frac{\partial^3 V}{\partial x^3}w_{ppp}\right] . \tag{33}$$

Read from the classical side, this is the correction that would make the centered-difference approximation exact; read from the quantum side, it is the leading Moyal correction [4, 17, 6] that separates Liouville from von Neumann dynamics. We note that the reading of this $\hbar^2$ term as the small parameter of a classical limit is not universally accepted: it has been argued that the correction series carries no explicit $\hbar$ and that the classical–quantum transition is set by spatial scale rather than by $\hbar \to 0$ [18]. The structural statement below does not depend on which reading one adopts.

The prefactor of this term is of order $\hbar^2$, but its size is not: it is small only for distributions with a short coherence length (Section 6.1), and its effects are secular: in anharmonic potentials the classical and quantum evolutions diverge over long times—classically by filamentation of the phase-space distribution, quantum mechanically by interference, recurrences, and wave-packet revivals. Where the two part company, experiment follows the quantum dynamics. The structural statement is that the *entire* difference between the dynamics of the two theories is one term, made exact in Section 6.1; for a quartic potential it is the single term above.

6.1 An exact form: the quantum tunneling correction

The error term above is the first term of an expansion, and the exact remainder has a closed form. Trivially,

$$[V(x+\hat{p}/2,t) - V(x-\hat{p}/2,t)]\,\hat{w} = \hat{p}\,\frac{\partial V}{\partial x}\,\hat{w} + \Delta\hat{Q}\,\hat{w}\,, \tag{34}$$

where

$$\Delta\hat{Q}(x,\hat{p},t) = [V(x+\hat{p}/2,t) - V(x-\hat{p}/2,t)] - \hat{p}\,\frac{\partial V}{\partial x}(x,t) = \int_{x-\hat{p}/2}^{x+\hat{p}/2} dy\left[\frac{\partial V}{\partial x}(y,t) - \frac{\partial V}{\partial x}(x,t)\right] \tag{35}$$

is the exact remaining correction of classical dynamics to quantum mechanics. We call it the *quantum tunneling correction*, after its most striking effect: without it, every trajectory is confined to its classical path, and a particle below a barrier stays on its side forever (the square potential below). Interference and zero-point structure are not its work: they are carried by w under classical flow, exactly so in a quadratic potential, where $\Delta\hat{Q} = 0$. In anharmonic potentials it also keeps eigenstates stationary and produces wave-packet revivals. Adding $\Delta\hat{Q}\,\hat{w}$ to the Fourier-transformed Liouville equation (24) yields (26) with no remainder:

$$i\hbar\hat{w}_t - \frac{\hbar^2}{2M}\left[\left(\frac{\partial}{\partial z_+}\right)^2 - \left(\frac{\partial}{\partial z_-}\right)^2\right]\hat{w} + \hat{p}\frac{\partial V}{\partial x}\hat{w} + \Delta\hat{Q}\,\hat{w} = 0\,. \tag{36}$$

The first three terms are classical mechanics and, simultaneously, the leading-order dynamics of quantum mechanics; $\Delta\hat{Q}$ is the remaining adjustment of classical dynamics to quantum.

The integral form in (35) makes the size of the adjustment clear. The exact force term averages $\partial V/\partial x$ over the segment from $x - \hat{p}/2$ to $x + \hat{p}/2$ joining the two branches; the classical term evaluates it once, at the midpoint. Classical mechanics is the midpoint rule for that integral, and $\Delta\hat{Q}$ is the midpoint rule's error. The linear part of $\partial V/\partial x$ integrates to zero over the symmetric segment, so $\Delta\hat{Q}$ sees only the curvature of the force, and the midpoint-rule error bound holds for every $\hat{p}$, not merely as the first term of a series:

$$|\Delta\hat{Q}(x,\hat{p},t)| \le \frac{|\hat{p}|^3}{24}\max_{|y-x|\le|\hat{p}|/2}\left|\frac{\partial^3 V}{\partial x^3}(y,t)\right|\,. \tag{37}$$

Expanding (35) about x recovers the error term above as its first term and, translated back to phase space, the Moyal series [4, 5]. The closed form is simpler than the series: in the $\hat{p}$ representation $\Delta\hat{Q}$ is an ordinary function, evaluated directly.

For a quadratic potential $\Delta\hat{Q} = 0$ identically, which is why the transform is exact there. For a quartic potential $V = x^4/4$, $\Delta\hat{Q} = x\hat{p}^3/4$ is exactly the leading term, and the $\hbar^2$ term is the whole difference. For a lattice $V = -V_0\cos kx$, $\Delta\hat{Q} = V_0\sin kx\,[\,2\sin(k\hat{p}/2) - k\hat{p}\,]$, which no finite number of Moyal terms reproduces. In three dimensions $\hat{p}$ is a vector and $\Delta\hat{Q} = V(x+\hat{p}/2) - V(x-\hat{p}/2) - \hat{p}\cdot\nabla V$.

Like $\hat{w}$, the correction $\Delta\hat{Q}$ lives in the Fourier representation, and it must be inverse-

transformed to act in phase space. The classical force term needs no such step beyond $\hat{p} \to -i\hbar\,\partial/\partial p$. The correction instead becomes a convolution in p, so that, exactly,

$$w_t + \frac{p}{M} w_x - \frac{\partial V}{\partial x} w_p = \int_{-\infty}^{+\infty} dq\; \Delta Q(x,q,t)\, w(x, p-q, t)\,, \tag{38}$$

$$\Delta Q(x,q,t) = \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar}\, e^{+i\hat{p}q/\hbar}\, \frac{i}{\hbar}\, \Delta\hat{Q}(x,\hat{p},t)\,. \tag{39}$$

Since $\Delta\hat{Q}$ is real and odd in $\hat{p}$, the kernel ΔQ is real and odd in q. It is local in x but spread over momentum transfers q, so it is not an ordinary Hamiltonian function $H(x,p)$ entering through a Poisson bracket: the exact dynamics is classical Liouville flow plus a nonlocal-in-momentum correction, not Liouville flow in a modified Hamiltonian. Because $\Delta\hat{Q}$ vanishes at $\hat{p} = 0$, $\int \Delta Q\, dq = 0$: the correction conserves probability, and a uniform background $w \to w + c$ remains invisible to it, as in Section 8.4. Section 6.3 reads (38) as classical trajectories with signed momentum jumps.

A square potential. For piecewise-constant V the classical force is an impulse at each edge: a trajectory below a barrier reflects, and one above it refracts. Away from the edges $\partial V/\partial x = 0$, so (35) is the whole difference of the potential across the segment,

$$\Delta\hat{Q}(x,\hat{p}) = V(x + \hat{p}/2) - V(x - \hat{p}/2)\,, \tag{40}$$

which vanishes until one branch point $z_\pm = x \pm \hat{p}/2$ reaches an edge (Figure 1a). A distribution whose coherence does not reach the walls feels only the classical impulse. For a single step of height s at x_j the kernel (39) is, for $x \neq x_j$,

$$\Delta Q(x,q) = -\frac{s}{\pi\hbar}\, \frac{\cos(2(x - x_j)\, q/\hbar)}{q}\,, \tag{41}$$

and a square well or barrier is a sum of such terms (Figure 1b). The kernel is odd in q, and both $\int \Delta Q\, dq$ and $\int q\, \Delta Q\, dq$ vanish away from the edges: it moves signed weight up and down in momentum while leaving the probability and the mean momentum unchanged. Near a wall it places weight at momenta above the barrier, and that weight then follows classical paths over the barrier and out. This is tunneling read from the classical side: free flight and classical reflection, plus momentum redistribution with zero mean.

The kernel itself does not fall off away from the barriers. For a thin barrier $V = \alpha\,\delta(x - x_0)$, $\Delta Q(x,q) = -(2\alpha/\pi\hbar^2)\sin(2(x_0 - x)q/\hbar)$: distance from the barrier sets how fast the kernel oscillates in q, with period $\pi\hbar/|x - x_0|$, not how large it is, and $\max_q |\Delta Q|$ is the same everywhere (Figure 1c). What localizes the correction is the distribution it acts on. A w

that is smooth on a momentum scale Δp averages away oscillations finer than Δp, which is the coherence-length statement above: smoothing the kernel over Δp multiplies $\Delta\hat{Q}$ by $e^{-\hat{p}^2\Delta p^2/2\hbar^2}$, and the smoothed kernel is confined to within about $\hbar/\Delta p$ of the barriers. The map of where tunneling matters is therefore a property of the kernel and a momentum resolution together, not of the kernel alone. Unsmoothed, Figure 1b suggests the simplest particle reading of the correction, a signed rate of momentum jumps q at each x; Section 6.3 develops it.

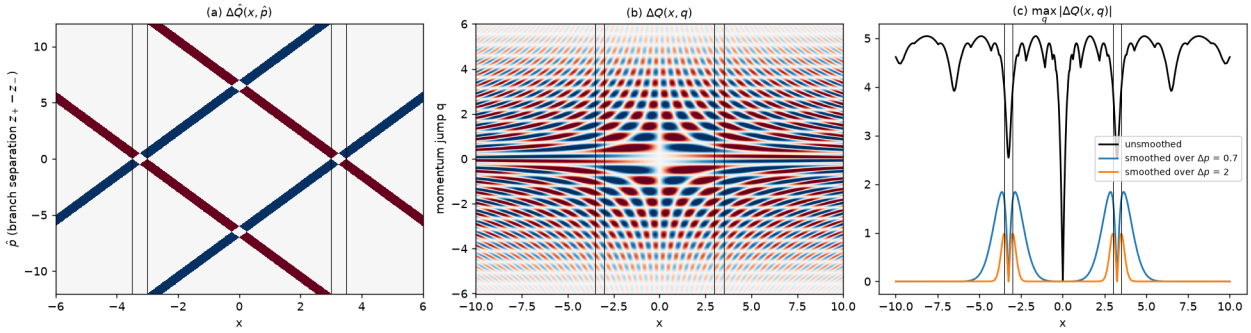


Figure 1: The quantum tunneling correction for a square well $|x| < 3$ between barriers $V_0 = 8$ of width 0.5 ($\hbar = 1$; edges marked). (a) $\Delta\hat{Q}(x,\hat{p})$ of (40), nonzero only where a branch point $x \pm \hat{p}/2$ lies in a barrier. (b) Its phase-space kernel $\Delta Q(x,q)$, the sum of (41) over the four edges: odd in q, with fringes that become finer, not weaker, away from the barriers. (c) $\max_q |\Delta Q(x,q)|$, unsmoothed (black) and smoothed over momentum resolutions $\Delta p = 0.7$, the width of the packet in Figure 2, and $\Delta p = 2$.

Figure 2 shows a Gaussian ($p_0 = 2$, energy well below the barrier) in a well $|x| < 3$ bounded by barriers of height $V_0 = 8$ and width 0.5 ($\hbar = M = 1$). Classical flow lets out only the 0.23% of the distribution that starts above the barrier; classical flow plus ΔQ lets out 32% by $t = 10$, matching exact quantum evolution to within 10^{-3} in the escaped probability.

The size of the correction is set by the coherence length rather than by $\hbar$ alone. The variable $\hat{p} = z_+ - z_-$ is the separation between the two branches, and $\Delta\hat{Q}$ is negligible when $\hat{w}$ is concentrated at $|\hat{p}|$ small compared with the length over which $\partial V/\partial x$ departs from linear. A smooth distribution of momentum width Δp has $|\hat{p}| \lesssim \hbar/\Delta p$ and sees almost the classical force. An energy eigenstate carries coherence across its entire classical orbit, so the correction is of order one at every quantum number, the semiclassical regime included. Filamentation narrows a classical distribution in p and so widens $\hat{w}$ in $\hat{p}$: the correction grows with time, which is the mechanism of the secular divergence described above.

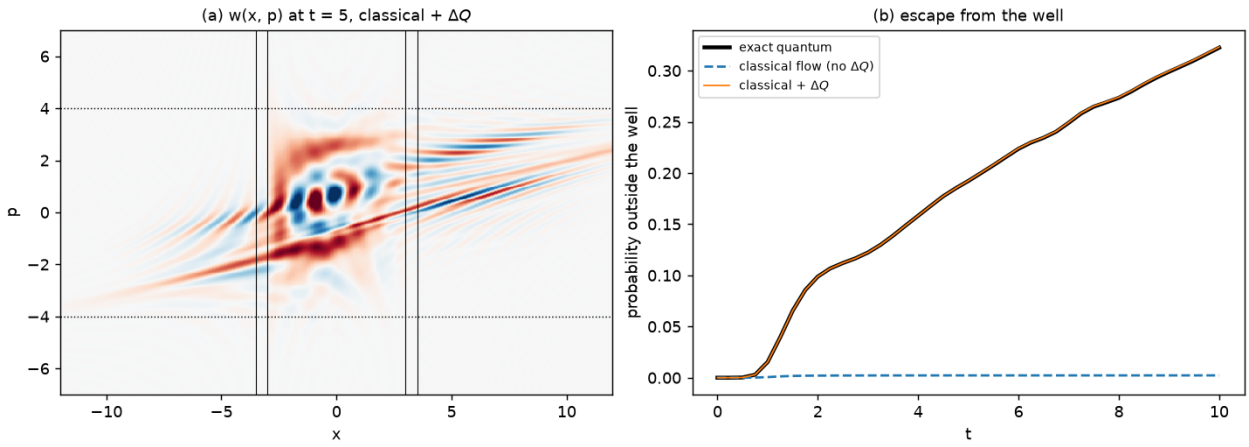


Figure 2: Escape from the square well of Figure 1 ($\hbar = M = 1$), starting from a Gaussian with $p_0 = 2$ below the barrier. (a) w at $t = 5$ under classical flow plus ΔQ; weight leaves the well at $|p| < \sqrt{2V_0}$ (dotted), which no classical trajectory can do. (b) Probability outside the well. Classical flow (dashed) stays trapped; classical flow plus ΔQ (orange) follows exact quantum evolution (black).

6.2 Efficient modeling via operator splitting

Equation (38) separates the exact dynamics into two parts,

$$w_t = \mathcal{C}\, w + \Delta\mathcal{Q}\, w\,, \qquad \mathcal{C}\, w = -\frac{p}{M}\, w_x + \frac{\partial V}{\partial x}\, w_p\,, \qquad \Delta\mathcal{Q}\, w = \int dq\; \Delta Q(x,q,t)\, w(x, p-q, t)\,, \tag{42}$$

where $\mathcal{C}$ is the classical Liouville operator, including the classical potential, and $\Delta\mathcal{Q}$ is the quantum tunneling correction. Each part is easy on its own. $e^{h\mathcal{C}}$ is classical mechanics: w is carried along classical trajectories, $w(x,p,t+h) = w(\Phi_{-h}(x,p),t)$, where the flow Φ comes from any ordinary differential equation solver for Hamilton's equations. $e^{h\Delta\mathcal{Q}}$ needs no solver at all: in $\hat{p}$, at fixed x, the correction is a multiplication, so $e^{h\Delta\mathcal{Q}}$ multiplies $\hat{w}$ by $\exp(ih\,\Delta\hat{Q}/\hbar)$ exactly. Strang splitting [19], with adjacent half-steps of $\mathcal{C}$ merged, advances w through a time $T = nh$ as

$$w(T) = e^{h\mathcal{C}/2}\left(e^{h\Delta\mathcal{Q}}\, e^{h\mathcal{C}}\right)^{n-1} e^{h\Delta\mathcal{Q}}\, e^{h\mathcal{C}/2}\; w(0) + O(h^2)\,. \tag{43}$$

The error per step is $h^3\left(\frac{1}{12}[\Delta\mathcal{Q},[\Delta\mathcal{Q},\mathcal{C}]] - \frac{1}{24}[\mathcal{C},[\mathcal{C},\Delta\mathcal{Q}]]\right) + O(h^5)$. Every term contains the commutator $[\mathcal{C},\Delta\mathcal{Q}]$, so the error is small wherever the correction is small, whatever the step size, and for a quadratic potential, where $\Delta\mathcal{Q} = 0$, the scheme is exact for any h. This is the natural factoring of the problem into classical mechanics plus quantum corrections. The usual split-step method for the Wigner function, the phase-space form of the split-step Fourier method for Schrödinger's equation [20], instead alternates free flight with the full

force, so it carries the commutator of free flight and classical force, which does not vanish even for a purely classical state.

We compared the two factorings in $V = x^4/4$ ($M = 1$) against exact quantum evolution in the eigenbasis at $t = 2T_0$, for step sizes from $T_0/800$ to $T_0/12.5$: the ground state and a displaced Gaussian at $\hbar = 1$, where the correction is of order one, and a coherent state at $\hbar = 0.1$. In (43) the classical flow $e^{h\mathcal{C}}$ was computed accurately, by classical substeps of $T_0/1600$; the correction was applied exactly. Both schemes converge at second order (Figure 3). At the same step size, away from the coarsest steps and the grid floor, factoring into classical flow and correction gives errors typically 2 to 3 times smaller in w and 2 to 5 times smaller in the Born density. The gain is modest rather than dramatic. For these pure states the correction is not small where the error accumulates, plausibly because its ∂_p^3 structure acts on the fine momentum structure that develops along the orbit. Larger gains are expected for states with short coherence lengths and for nearly quadratic potentials, which we have not tested. The practical advantage of (43) is structural: the classical step can use any accurate trajectory integrator, including large adaptive steps or exact orbits, and the only quantum work per step is one Fourier transform in p and one multiplication.

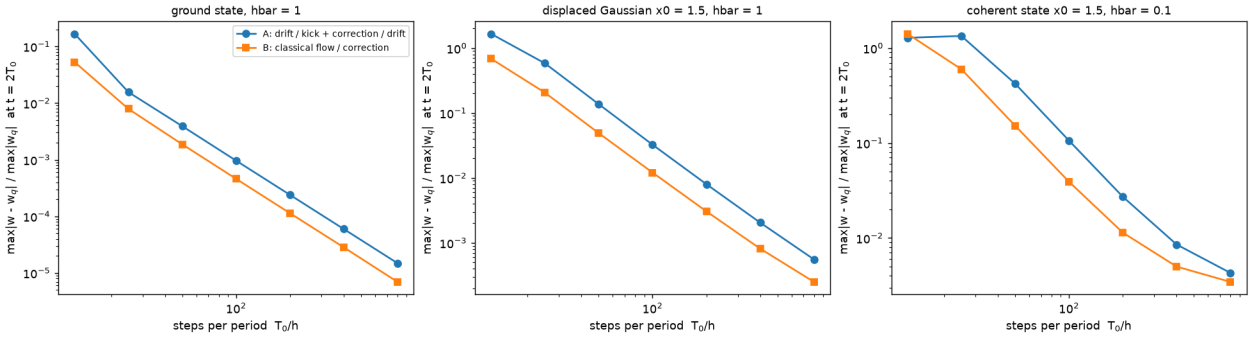


Figure 3: Error of operator splitting in $V = x^4/4$ at $t = 2T_0$ against exact quantum evolution, versus steps per classical period. A: the usual split of free flight and full force. B: the split (43) into classical flow $e^{h\mathcal{C}}$, computed accurately, and the exact correction $e^{h\Delta\mathcal{Q}}$. Both are second order; B is 2–3 times more accurate at the same step. The $\hbar = 0.1$ panel levels off near 3×10^{-3} at the grid resolution.

6.3 Classical trajectories with signed momentum jumps

Away from the edges of a square potential the classical force vanishes, so ΔQ is the entire force, and $\Delta Q(x, q)\, dq\, dt$ is the signed weight sent from momentum p to $p + q$ (Figure 1b). Nothing in that reading needs smoothing: sampled as jumps, the kernel reproduces the exact dynamics on average. We now develop it for a general potential. Section 6.1 kept the

classical force and added a correction. Returning the *whole* force term of (26) to phase space gives a simpler picture: particles fly freely between events, and the force acts by moving signed weight across finite momentum transfers. In (26) the force enters as the multiplier $\hat{w}_t = (i/\hbar)[V(z_+,t) - V(z_-,t)]\,\hat{w} + \cdots$. A product in $\hat{p}$ is a convolution in p, so, with nothing dropped,

$$w_t + \frac{p}{M}\,w_x = \int_{-\infty}^{+\infty} dq\;\Theta(x,q,t)\,w(x,p-q,t)\,, \tag{44}$$

$$\Theta(x,q,t) = \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar}\,e^{+i\hat{p}q/\hbar}\,\frac{i}{\hbar}\left[V(x+\hat{p}/2,t) - V(x-\hat{p}/2,t)\right]\,. \tag{45}$$

This is the integral form of the Wigner equation [2, 5]. The bracket is real and odd in $\hat{p}$, so Θ is real and odd in q. The left side of (44) is classical free flight; the right side replaces the classical force term $(\partial V/\partial x)\,w_p$, which corresponds to the kernel $(\partial V/\partial x)\,\delta'(q)$. The difference between the two kernels is the correction $\Delta Q(x,q)$ of (38).

Read (44) as a rule for particles. Because Θ is odd, its effect comes in pairs: in a time dt, a particle at (x,p) produces a copy of weight $+\Theta(x,q)\,dq\,dt$ at $p+q$ and a copy of the opposite weight at $p-q$, for each $q>0$. The moments of the kernel say what these pairs do:

$$\int \Theta\,dq = 0\,, \qquad \int q\,\Theta\,dq = -\frac{\partial V}{\partial x}\,, \qquad \int q^2\,\Theta\,dq = 0\,, \qquad \int q^3\,\Theta\,dq = \frac{\hbar^2}{4}\frac{\partial^3 V}{\partial x^3}\,. \tag{46}$$

The pairs create no probability. On average they deliver exactly the classical force, and at each x they change the first two momentum moments of w exactly as the classical force does. They first differ from the classical force at the third moment; expanding $w(x,p-q)$ in powers of q recovers the Moyal series [4] term by term. The motion between events is classical and the average push is classical; what is not classical is that weight branches into signed pairs, so this is not a Liouville flow along a Hamiltonian vector field.

A lattice: discrete jumps. For a single Fourier mode the kernel is a pair of delta functions. For $V = -V_0\cos kx$,

$$w_t + \frac{p}{M}\,w_x = \frac{V_0\sin kx}{\hbar}\left[w\big(x,p+\tfrac{\hbar k}{2},t\big) - w\big(x,p-\tfrac{\hbar k}{2},t\big)\right]\,. \tag{47}$$

The force term is a centered difference in momentum with step $\hbar k$, the counterpart of the centered difference in position (14) on which the looking-glass transform is built, and it becomes $(\partial V/\partial x)\,w_p$ as $\hbar k \to 0$. Every jump is exactly $\pm\hbar k/2$, and a general potential contributes such a pair of jumps for each of its Fourier components. The half-step is a feature of w, not of measured momenta: w places interference terms midway between momentum components, and these integrate to zero over x.

Polynomial potentials: Airy jumps. For a polynomial potential Θ is a finite sum of derivatives of $\delta(q)$, and (44) is the terminating Moyal equation. The jumps then appear over a finite time step Δt. For $V = x^4/4$, the classical kick plus the residual multiplier $\exp(i\,\Delta t\,\Delta\hat{Q}/\hbar) = \exp(i\,\Delta t\,x\hat{p}^3/4\hbar)$ is exact, and the residual's kernel is an Airy function,

$$K_{\Delta t}(x,q) = \frac{1}{|c|}\,\mathrm{Ai}\Big(\frac{q}{c}\Big)\,, \qquad c = \left(\frac{3\hbar^2 x\,\Delta t}{4}\right)^{1/3}\,. \tag{48}$$

Its width grows as $(\hbar^2\Delta t)^{1/3}$: the redistribution is neither a drift nor a diffusion.

Numerical check. We tested (44) against exact quantum evolution in an eigenbasis ($\hbar = M = 1$), applying the jump kernel deterministically to w on a phase-space grid. The classical baseline evolves the same initial w without jumps. In the quartic well $V = x^4/4$, over six classical periods T_0, we evolved the ground state and a Gaussian of width 0.6 displaced to $x = 1.5$. In the lattice $V = -20\cos 2\pi x$, we evolved a Gaussian of width 1.5 sites at rest on a well, in a periodic box of 64 sites; there the classical baseline is 4×10^5 sampled trajectories, because classical filaments outrun any fixed momentum grid. Table 1 reports the total-variation distance between the Born densities $\int w\,dp$ and the exact $|\Psi|^2$. Figure 4 shows the quartic case. In the lattice, 97% of the classical particles start below the barrier and stay trapped in their wells; with the jumps the packet spreads through the lattice by tunnelling, and its width follows the exact quantum width (Figure 5). Since (44) is an identity, agreement is guaranteed up to discretization. The tests check the decomposition, and they measure how much of the dynamics the jumps carry: all of the difference between classical and quantum evolution.

Born-density distance from exact quantum	classical flow	trajectories + jumps
Quartic ground state, $t \le 6T_0$	0.063	$< 10^{-5}$
Quartic displaced Gaussian, $t \le 6T_0$	0.31–0.50	$\le 1.6 \times 10^{-3}$
Lattice Gaussian, $0.3 \le t \le 3$	0.17–0.34	$\le 3.4 \times 10^{-4}$

Table 1: Total-variation distance $\frac{1}{2}\int |\rho - |\Psi|^2|\,dx$ between the Born density of the evolved w and exact quantum evolution. Classical flow drops the jumps of (44); trajectories + jumps keep them.

Particles and the sign problem. This reading is the basis of signed-particle Monte Carlo methods for Wigner transport in semiconductor devices [21], where particles follow classical trajectories and the kernel Θ (there called the Wigner potential) creates $\pm$ particle pairs. Sellier promoted that method to a formulation of non-relativistic quantum mechanics in its own right: quantum systems as ensembles of field-less classical particles that carry a sign and interact with the potential only through creation and annihilation events [22]. Here the same

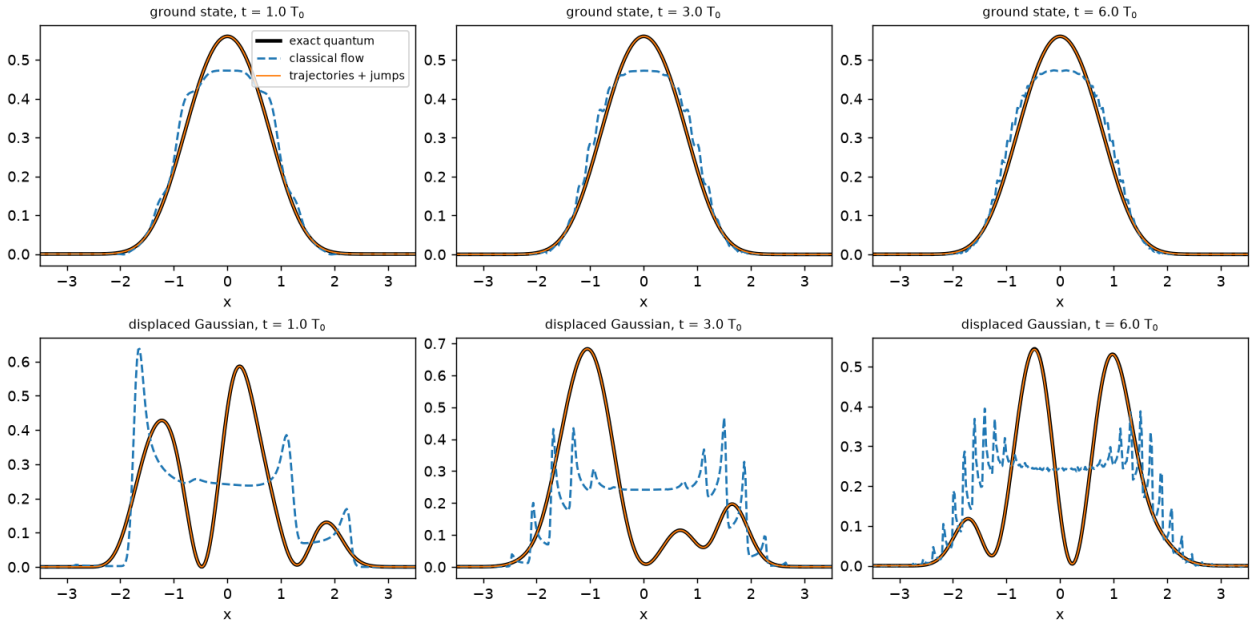


Figure 4: Born densities in $V = x^4/4$ ($\hbar = M = 1$) at $t = T_0$, $3T_0$ and $6T_0$. Top: the ground state, which is stationary. Bottom: a displaced Gaussian. Exact quantum evolution (black), classical flow (dashed), and classical trajectories with the signed jumps of (44) (orange). Classical flow filaments; the jumps restore the interference pattern.

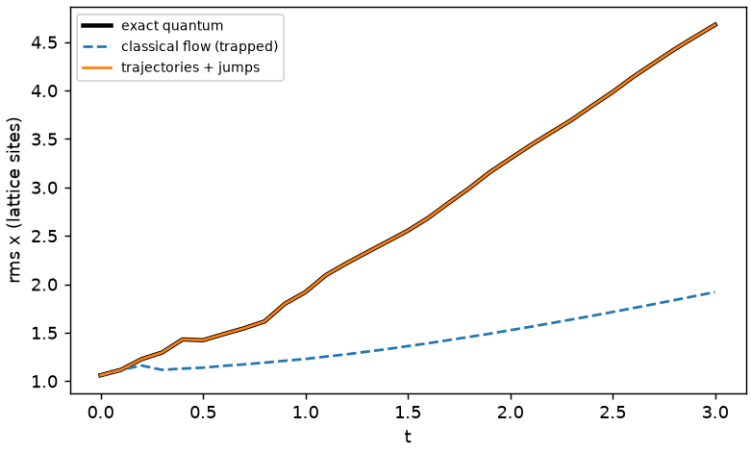


Figure 5: Root-mean-square width of a Gaussian packet in the lattice $V = -20\cos 2\pi x$ ($\hbar = M = 1$, width in lattice sites). Classical trajectories stay mostly trapped in their wells; classical trajectories with the jumps of (47), each exactly $\pm\hbar k/2$, spread by tunnelling and track exact quantum evolution.

picture arises from the classical side, as the phase-space form of the exact correction $\Delta\hat{Q}$, and (47) shows its simplest case: in a lattice the creation events are a centered difference. Sampled one particle at a time, the pairs carry a cost: each pair leaves $\int w$ unchanged but adds $2|\Theta|\,dq\,dt$ to $\int |w|$, so the total absolute weight grows exponentially unless positive and negative copies are allowed to annihilate. In our quartic test, stochastic pairs drove the total absolute weight from 1.2 to 5 within 50 time steps and to 5×10^5 within 200. The square well is the sharp case. There the rate $\gamma(x) = \int |\Delta Q(x,q)|\,dq$ does not depend on the distance to the barriers (Figure 1c), and it grows logarithmically with the largest jump allowed, from 16 for $|q| < 8$ to 35 for $|q| < 256$. Every particle in the well jumps tens of times per unit time, and tunneling emerges only from near-complete cancellation between signs; the absolute weight grows like $e^{\gamma t}$, about 5×10^8 by $t = 1$ for $|q| < 16$, the time at which the packet first reaches the wall. Applied deterministically, as above, the same kernel has no such growth. The exact dynamics is therefore classical in its trajectories and in its mean force, with the quantum content carried by signed branching of the weight.

7 The Born Projection

The standard Born rule states that at time t, the probability density of observing position x is $|\Psi(x,t)|^2$. We recover this as the instantaneous marginal of w over momentum, which we call the Born projection:

$$
\begin{aligned}
\text{born}\, w &= \int_{-\infty}^{+\infty} dp\, w(x,p,t)\,, && (49)\\
&= \int_{-\infty}^{+\infty} dp \int_{-\infty}^{+\infty} \frac{d\hat{p}}{2\pi\hbar}\, e^{i\frac{\hat{p}p}{\hbar}}\, \hat{w}\,, && (50)\\
&= \int_{-\infty}^{+\infty} d\hat{p}\, \hat{w} \int_{-\infty}^{+\infty} \frac{dp}{2\pi\hbar}\, e^{i\frac{\hat{p}p}{\hbar}}\,, && (51)\\
&= \int_{-\infty}^{+\infty} d\hat{p}\, \hat{w}\, \delta(\hat{p})\,, && (52)\\
&= \hat{w}|_{\hat{p}=0}\,, && (53)\\
&= \sum_{\lambda,\lambda'} C_{\lambda,\lambda'} e^{i(\lambda-\lambda')t} \psi^*_\lambda(x)\psi_{\lambda'}(x)\,, && (54)\\
&= \sum_k c_k \sum_{\lambda,\lambda'} (a_k)^*_\lambda (a_k)_{\lambda'} e^{i(\lambda-\lambda')t} \psi^*_\lambda(x)\psi_{\lambda'}(x)\,, && (55)\\
&= \sum_k c_k \left| \sum_\lambda (a_k)_\lambda e^{-i\lambda t} \psi_\lambda(x) \right|^2 . && (56)
\end{aligned}
$$

This is the standard Born position density, obtained here as the momentum marginal of the phase-space distribution. Marginal-based presentations of the Born densities are standard in the phase-space formulation [6, 23], and the claim that they are thereby *derived* appears in [7]. Within this representation the Born rule for position is not an additional postulate: it is ordinary marginalization, and no collapse dynamics or decoherence mechanism [17] is invoked to obtain the statistics.

We are explicit about the limits of this deflation. It reproduces the statistics of position (and, by the symmetric argument, momentum) measurements. It does not by itself explain definite individual outcomes or the statistics of observables that are not phase-space marginals—those require overlaps $\int w\, W_A$ with Wigner symbols W_A that themselves take negative values. The update of w conditioned on an observed result, for measurements that are marginals, is treated in Section 8.3. The remaining gaps are collected in Section 9.

8 Interpretive Remarks

8.1 Heisenberg's Uncertainty Principle

Within the looking-glass dynamics there is nothing inconsistent about a distribution that specifies the precise initial position and momentum of the particle; the equations evolve it without complaint. However, we can only observe the marginal over momentum, which makes the view fuzzy.

Substitution of $w_0(x,p) = \delta(x-x_0)\delta(p-p_0)$ into the formula for $C_{\lambda,\lambda'}$ gives the cross-Wigner function as the coefficients of the expansion of $\hat{w}$ in the eigenbasis of the Schrödinger equation:

$$C_{\lambda,\lambda'}[\delta(x-x_0)\delta(p-p_0)] = \int_{-\infty}^{+\infty} d\hat{p}\, \psi_\lambda(x_0+\frac{\hat{p}}{2})\psi^*_{\lambda'}(x_0-\frac{\hat{p}}{2})e^{-i\frac{\hat{p}p_0}{\hbar}}\,. \tag{57}$$

For a symmetric potential V, the eigenfunctions ψ_λ are either symmetric or antisymmetric, $\psi_\lambda(-x) = \sigma_\lambda\psi_\lambda(x)$, where $\sigma_\lambda = \pm 1$. We can use this to compute $C_{\lambda,\lambda'}$ for the $x_0 = 0$,

$p_0 = 0$ case:

$$
\begin{aligned}
C_{\lambda,\lambda'}[\delta(x)\delta(p)] &= \int_{-\infty}^{+\infty} d\hat{p}\, \psi_\lambda(+\frac{\hat{p}}{2})\psi^*_{\lambda'}(-\frac{\hat{p}}{2})\,, && (58)\\
&= \sigma_{\lambda'} \int_{-\infty}^{+\infty} d\hat{p}\, \psi_\lambda(\frac{\hat{p}}{2})\psi^*_{\lambda'}(\frac{\hat{p}}{2})\,, && (59)\\
&= 2\sigma_{\lambda'} \int_{-\infty}^{+\infty} du\, \psi_\lambda(u)\psi^*_{\lambda'}(u)\,, && (60)\\
&= 2\sigma_\lambda \delta_{\lambda,\lambda'}\,. && (61)
\end{aligned}
$$

The Born projection yields

$$
\mathrm{born}\,\delta(x)\delta(p) = 2\sum_\lambda \sigma_\lambda |\psi_\lambda(x)|^2\,. \tag{62}
$$

This is the expansion of $\delta(x)$ as a distribution—what we would expect for a particle at the bottom of the well with no momentum.

Two caveats sharpen this picture. First, the coefficient matrix C of the point preparation is Hermitian with unit trace but *not* positive semidefinite: the eigenvalues $c_k = 2\sigma_\lambda$ alternate in sign. A point in phase space is therefore not a quantum state—necessary and sufficient conditions for a phase-space function to be a Wigner distribution are known [12]—and the uncertainty principle reappears in this representation as the positivity condition on C, not merely as fuzziness of the marginals. Second, the looking-glass dynamics is nevertheless well posed on the larger set of Hermitian unit-trace C, and positivity is preserved by the evolution, so the quantum states form an invariant subset. Classical point preparations live in the complement: they evolve consistently and project to sensible signed densities, as above, but no quantum preparation reaches them. What principle selects the positive subset is an open question we return to in Section 9.

8.2 Quantum Entanglement

The above derivation is for a single particle in a single space dimension, but it is trivial to extend to multiple particles in three spatial dimensions. Entanglement amounts to a dependent choice of initial distribution, so $\hat{w}$ cannot be factored into independent components for each particle. The underlying mechanism evolves independently; it is only the initial distribution and the momentum marginal that make it look like there is something mysterious. It is a consequence of dependent distributions and the projection.

For two particles with single-particle eigenstates $\hat{\alpha} = \psi_\lambda$ and $\hat{\beta} = \psi_\mu$, the coefficient

matrix C extends to the two-particle eigenbasis $\{|\alpha\alpha\rangle, |\alpha\beta\rangle, |\beta\alpha\rangle, |\beta\beta\rangle\}$. The four maximally entangled Bell states are pure states $C = a\,a^\dagger$ with

$$a_{\Phi^\pm} = \tfrac{1}{\sqrt{2}}(|\alpha\alpha\rangle \pm |\beta\beta\rangle)\,, \tag{63}$$

$$a_{\Psi^\pm} = \tfrac{1}{\sqrt{2}}(|\alpha\beta\rangle \pm |\beta\alpha\rangle)\,. \tag{64}$$

None of these factor as $|s_1\rangle \otimes |s_2\rangle$; entanglement is precisely this non-factorizability of a. The outer product $a\,a^\dagger$ generates off-diagonal coherence terms in $\hat{w}$ that distinguish entangled states from classical mixtures, which would have C diagonal.

This demystification of entanglement—as non-factorizable statistical dependence in the joint initial data—is not new in spirit: related readings appear in Wetterich's derivation of entanglement from classical statistics [24] and in Ozorio de Almeida's representation of an entangled state as a non-factorizable Lagrangian surface in phase space [25]. It also has a sharp limit that must be stated. A Bell-type entangled state admits *no* non-negative joint phase-space distribution reproducing all of its measurement statistics; this is a finite, explicit no-go [26, 27]. The dependent object here is therefore the signed w of Section 8.4, not an ordinary non-negative probability: entanglement is statistical dependence of initial data in that signed field. The obstruction is moreover observable-dependent—a positive-Wigner Gaussian state still violates a Bell inequality when displaced-parity observables are measured [28], precisely because those observables are not phase-space marginals. The remark above thus demystifies the *kinematics* of entanglement (non-factorizable C); measurement and conditioning are taken up next.

8.3 Collapse of a Separated Entangled Pair

The kinematic reading above leaves the sharpest question open: what happens to A when a distant B is measured? We model measurement in the von Neumann manner [29], as interaction with an apparatus, and take the interaction to involve B alone. For this question no relativity is needed, and the particles may be free.

Every step is exact. The error term of Section 6 involves $\partial^3 V/\partial x^3$, and more generally the Moyal correction vanishes for any Hamiltonian at most quadratic in the phase-space variables. For two free particles the looking-glass dynamics is therefore exactly the classical free streaming of the joint distribution,

$$w(x_A, p_A, x_B, p_B, t) = w_0\!\left(x_A - \frac{p_A t}{M},\, p_A,\; x_B - \frac{p_B t}{M},\, p_B\right)\,, \tag{65}$$

and nothing quantum happens dynamically: everything quantum about the pair resides in w_0, in its correlations and its sign. Couple B to a pointer with position X and momentum P by an impulsive interaction $H_{\rm int} = g(t)\, x_B P$ of integrated strength g. It is bilinear, so the centered difference remains exact, and Hamilton's equations give

$$X \to X + g\, x_B\,, \qquad p_B \to p_B - gP\ : \tag{66}$$

the pointer records B's position, and B receives a momentum kick. Liouville flow carries the joint distribution to

$$w_{\rm after} = w_{AB}(x_A, p_A, x_B, p_B + gP)\; w_{\rm ptr}(X - g x_B, P)\,. \tag{67}$$

The interaction involves only B and its apparatus; integrating out B and the pointer leaves A's marginal unchanged. The flow is local.

Collapse is conditioning. Reading the pointer is taking its X-marginal, the same operation as the Born projection. Conditioning on the reading $X = X_0$ and integrating out P and p_B—which removes the kick—gives

$$w_A(x_A, p_A \,|\, X_0) \;\propto\; \int dx_B\; \rho_{\rm ptr}(X_0 - g x_B) \int dp_B\; w_{AB}(x_A, p_A, x_B, p_B)\,, \tag{68}$$

where $\rho_{\rm ptr}$ is the pointer's initial position density. This is exactly the Wigner function of A's quantum state after an unsharp position measurement on B; in the sharp-pointer limit it reduces to the standard identity that the Wigner function of $\langle x_B|\hat{\rho}|x_B\rangle$ is the joint w marginalized over p_B and evaluated at x_B. Averaging (68) over readings returns A's original marginal: nothing done at B signals to A. The instantaneous change in A's description is a change of conditioning, not a change of dynamics. Within this representation, the "collapse" of a separated entangled pair is conditioning of a joint distribution on the marginal of a pointer that interacted only with B.

Where the quantum ingredient lives. If $w_{AB} \geq 0$, (68) is ordinary classical conditioning; Bell made exactly this point for the original EPR state, whose Wigner function is non-negative and is itself a local hidden-variable model [13]. If w_{AB} is signed, (68) is formally identical but is Bayes' rule with signed weights. Different measurement choices—measuring x_B at time t is measuring the quadrature $x_B - p_B t/M$ at time 0—are different marginals of the same signed w, and no single non-negative distribution reproduces them all; non-Gaussian states violate Bell inequalities with such quadrature measurements alone

[30, 31]. The conditional is a signed field divided by a positive marginal probability, so it is an object of the same kind as w itself, and its readings are again non-negative marginals.

The size of the apparatus. An apparatus of N pointer modes, each coupled as $g\,x_B P_k$ with initial width σ, reduces exactly to one collective pointer: $\bar{X} = N^{-1}\sum_k X_k$ and $P_{\mathrm{tot}} = \sum_k P_k$ are conjugate, the coupling involves only P_{tot}, and the $N-1$ relative modes decouple. The collective pointer is a minimum-uncertainty Gaussian of width $\sigma/\sqrt{N}$, so a large apparatus is one whose collective reading is sharp even when no single mode can resolve B. The price is the uncertainty principle as back-action: the kick gP_{tot} has variance $Ng^2\hbar^2/4\sigma^2$. In the momentum distribution this kick is a convolution, and an A–B interference term oscillating in p_B at the conjugate separation Δx_B is suppressed by

$$\exp\!\left(-\frac{Ng^2\Delta x_B^2}{8\sigma^2}\right). \tag{69}$$

In the larger state space of Section 8.1, a point pointer $\delta(X)\delta(P)$ would measure B perfectly without disturbing it; the positivity condition on the pointer is what forces the back-action.

Numerical illustration. We take $\hbar = M = 1$, Gaussian packets $\phi_c(x)$ of position width 0.5 centered at c, and the entangled state

$$\Psi_{AB} = \alpha\,|\mathrm{cat}_+\rangle_A|L\rangle_B + \beta\,|\mathrm{cat}_-\rangle_A|R\rangle_B\,, \qquad |\alpha|^2 = 0.7\,, \tag{70}$$

with $\mathrm{cat}_\pm \propto \phi_{-2} \pm \phi_{+2}$ and $L, R = \phi_{\mp 2}$. The apparatus couples only to B with $g = 1$; a single pointer of width $\sigma = 6$ cannot resolve B's branches, which are $\Delta x_B = 4$ apart. The classical calculation never uses a wavefunction after constructing w_0: it pushes signed phase-space weights through the Hamiltonian flow above, reads the collective pointer, and conditions on the sign of the reading r. An independent quantum calculation evolves $\Psi(x_A, x_B, X)$ on a grid with $\exp(-ig\,x_B\hat{P}/\hbar)$ and projects. The code and the environment that runs it accompany the paper.

Figures 6 and 7 and Table 2 show five features. The probability of each reading approaches the Born weight as the apparatus grows. A's unconditional distribution is unchanged by the measurement, and the probability-weighted average of the two conditionals reproduces it to machine precision. Conditioning selects A's branch: given $r < 0$, A is cat_+, and given $r > 0$, it is cat_-. The apparatus destroys the A–B coherence at the rate (69), after which the joint momentum distribution is indistinguishable from a classical mixture of the branches. And conditioning *creates* negativity: A's minimum falls from -0.09 unconditionally to $-1/\pi\hbar \approx -0.318$ given the reading, which conditioning with non-negative weights

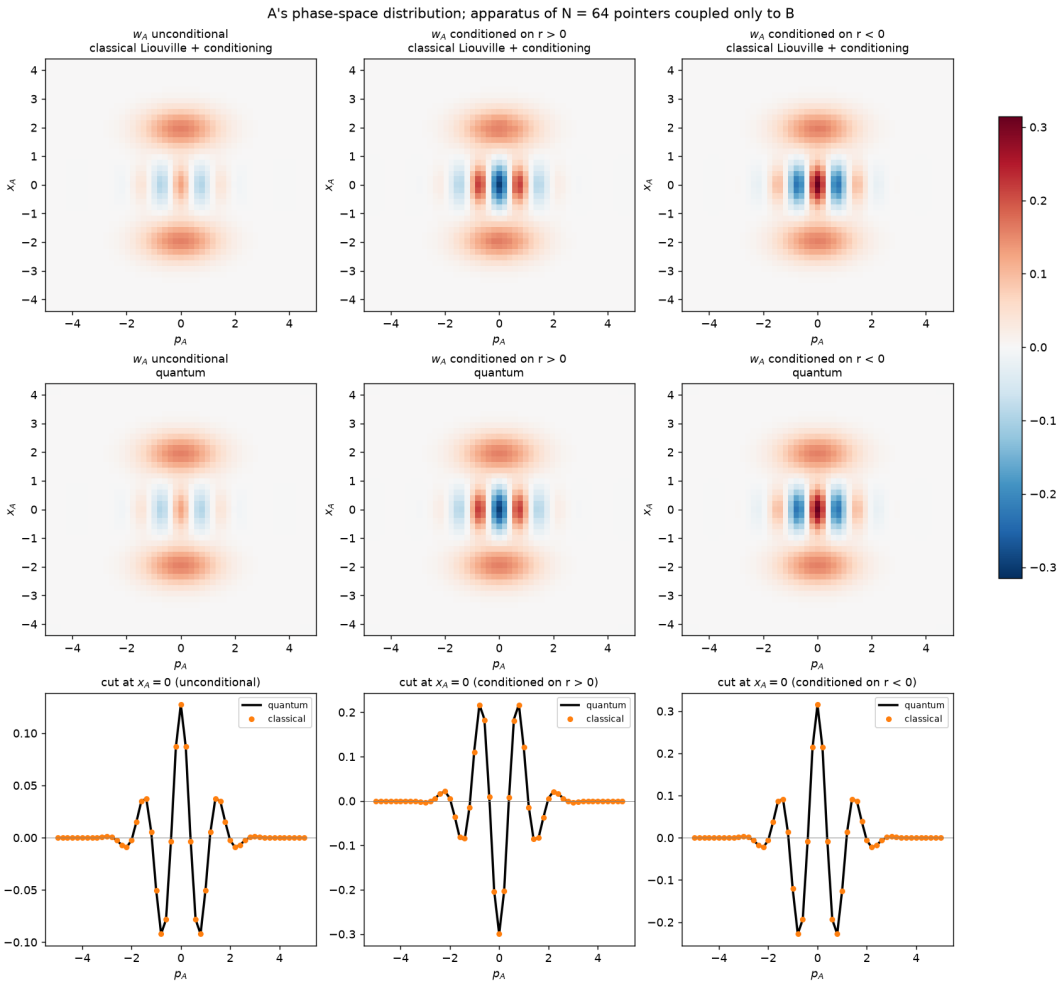


Figure 6: A's phase-space distribution for $N = 64$ pointers coupled only to B. Top row: classical Liouville flow of the signed w_{AB} followed by conditioning on the pointer reading. Middle row: independent quantum calculation. Bottom row: cuts at $x_A = 0$ (line: quantum; dots: classical). Conditioning on $r < 0$ or $r > 0$ selects cat_+ or cat_-, with $w_A(0,0) = \pm 1/\pi\hbar$, the bound of Section 8.4.

N	$P(r>0)$		A–B coherence		$\min w_A(\cdot \mid r>0)$	
	classical	quantum	classical	quantum	classical	quantum
1	0.448	0.446	0.96	0.95	−0.040	−0.036
4	0.403	0.400	0.81	0.80	−0.035	−0.036
16	0.342	0.340	0.41	0.41	−0.185	−0.185
64	0.306	0.305	0.03	0.03	−0.300	−0.299
256	0.300	0.300	0.02	0.00	−0.318	−0.317

Table 2: Classical (signed Liouville flow plus conditioning) against quantum results. The coherence column for (69) reads 0.95, 0.80, 0.41, 0.03, 0.00; the classical floor near 0.02 is Monte Carlo noise from the signed weights.

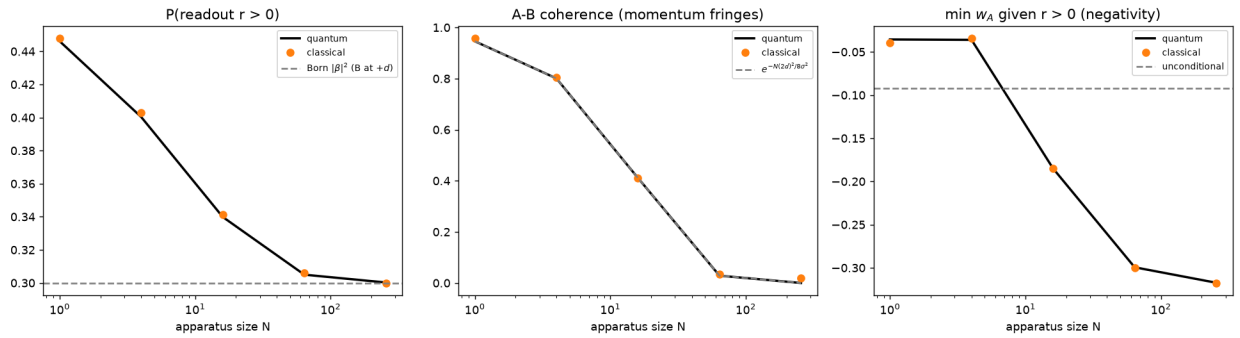


Figure 7: Dependence on apparatus size N. Left: probability of the reading $r > 0$, approaching the Born weight $|\beta|^2 = 0.3$. Center: A–B coherence, measured by the interference fringe of the joint momentum distribution, against (69). Right: minimum of A's conditional distribution. Line: quantum; dots: classical.

could never do.

Three limits should be stated. First, because every Hamiltonian here is quadratic, agreement between the classical and quantum calculations is guaranteed; it verifies the construction and the code, not a prediction that could have failed. The content is the reading—local flow, collapse as conditioning, Born weights from a pointer marginal—not the agreement. Second, "large" enters only through the sharpness of the collective reading; the record here is reversible in principle, and amplification into an irreversible record is not modeled. Third, the argument covers measurements that are phase-space marginals; observables that are not (Section 9) are outside it.

8.4 Negative Values of w

Every reader meeting a phase-space distribution that dips below zero asks the same question, so we answer it directly: the negative values do not matter, for two independent reasons.

w is a field, like a pressure wave. The Liouville equation (19) is a statement about *flux*, not about absolute values. If w satisfies it, so does $w + c$ for any constant c: a uniform background has zero gradient and carries no flux. An acoustic pressure wave is the familiar example. The pressure perturbation is negative in every rarefaction, and nobody objects, because what is physical is the total pressure, ambient level plus perturbation, and only gradients drive the flow. The same holds here. For every pure state the Wigner function obeys

$$-\frac{1}{\pi\hbar} \le w(x,p,t) \le \frac{1}{\pi\hbar}\,, \tag{71}$$

so $w = \rho - c$ with $\rho \geq 0$ and the universal ambient level $c = 1/(\pi\hbar)$, one count per quantum cell. The negative regions of w are rarefactions below that level, not places where probability is missing. Like the ambient pressure of an unbounded medium, the background has no finite total, and it needs none: it carries no dynamics and enters no prediction. We have not found this reading in the negativity literature, where negativity is usually treated as the irreducibly quantum feature [11].

A correct model owes us nothing more. A model is judged by its predictions, and every prediction made here is non-negative. The Born projection $\mathrm{born}\, w = \int dp\, w = |\Psi(x,t)|^2$ and the momentum marginal $\int dx\, w = |\tilde{\Psi}(p,t)|^2$ are genuine probability densities whatever the sign of w, as are the pointer readings of Section 8.3. Intermediate quantities in a calculation need not be probabilities; Feynman made this argument for negative probabilities in general [32]. With the correction $\Delta\hat{Q}$ of Section 6.1 the model is exact, and that is the only test that applies to it; flow in the bare potential is its approximation, accurate where the error term of Section 6 is negligible. None of this has anything to do with sign.

What the negative regions mark. Negative regions have been measured directly by quantum state tomography [33]. By Hudson's theorem [34], $w \geq 0$ everywhere exactly for Gaussian states, so the rarefactions mark the states with no Gaussian description, such as Fock states and superpositions. They are also where tunneling and interference show up in phase space; those effects are carried by the retained Schrödinger terms, not by the Moyal correction that is dropped for anharmonic potentials.

8.5 Many Worlds

The many-worlds interpretation [1] posits branching universes to resolve the measurement problem: if the wavefunction never collapses, what selects the observed outcome? The present framework removes one motivation for branching: the Born statistics arise by marginalization, with no collapse event anywhere in the dynamics. The apparent indeterminacy of quantum mechanics enters through the Born projection—the momentum marginal—which discards momentum information that is inaccessible to a position measurement, identical in form to how a classical observer recovers a position probability from an (x,p) distribution. This does not by itself resolve the measurement problem—Section 7 lists what marginalization leaves unexplained—but it relocates the apparent indeterminacy from the dynamics to the projection, where classical intuition can engage it.

9 Further Directions

Several questions remain open, each a direction for further work.

9.1 Spin and statistics

The construction covers spinless particles. Antisymmetric a vanishes for coincident single-particle states, so Pauli exclusion is representable, but nothing in the transform explains why identical particles should carry symmetric or antisymmetric C; spin and the spin-statistics connection are absent.

9.2 General observables

The Born densities for position and momentum are marginals of w. A general observable A requires the overlap $\int dx\, dp\, w\, W_A$, where the Wigner symbol W_A itself takes negative values, and the clean reading of measurement as marginalization is then lost. Whether the looking-glass framing offers any purchase on general measurements is open.

9.3 The larger state space

The dynamics is well posed for every Hermitian, unit-trace C; positivity is preserved by the evolution; and the quantum states form an invariant subset whose complement contains all classical point preparations. Positivity, like normalization, is a property of the initial data that the evolution preserves, so which subset nature occupies is a question about preparation—a prior—rather than about dynamics. Two questions follow. Do non-positive C—classically preparable in this representation—have any operational meaning, or are they strictly bookkeeping? And does the boundary of the quantum subset, where some $c_k = 0$, play a distinguished dynamical role? The conditions of [12] give the technical footing for these questions.

9.4 The uniform background

The ambient level $c = 1/(\pi\hbar)$ makes $\rho = w + c \geq 0$ for every pure state. Like the ambient pressure of an unbounded medium it carries no dynamics; whether it acquires a physical role in a finite box, or in field theory as vacuum fluctuation, is unexplored.

9.5 Pedagogy

The route from Liouville to Schrödinger requires only Fourier transforms and centered differences—tools familiar to any student of numerical methods. A classroom treatment with numerical experiments (exactness for the harmonic oscillator, visible error growth for quartic potentials, revivals versus filamentation) would test whether the looking-glass framing actually transfers classical intuition to the quantum domain.

References

[1] Bryce DeWitt and R. Neill Graham, editors. *The Many-Worlds Interpretation of Quantum Mechanics*. Princeton Series in Physics. Princeton University Press, Princeton, New Jersey, 1973.

[2] E. P. Wigner. On the quantum correction for thermodynamic equilibrium. *Physical Review*, 40:749–759, 1932.

[3] H. J. Groenewold. On the principles of elementary quantum mechanics. *Physica*, 12(7):405–460, 1946.

[4] J. E. Moyal. Quantum mechanics as a statistical theory. *Mathematical Proceedings of the Cambridge Philosophical Society*, 45:99–124, 1949.

[5] M. Hillery, R. F. O'Connell, M. O. Scully, and E. P. Wigner. Distribution functions in physics: Fundamentals. *Physics Reports*, 106(3):121–167, 1984.

[6] W. B. Case. Wigner functions and Weyl transforms for pedestrians. *American Journal of Physics*, 76(10):937–946, 2008.

[7] E. Carnovali Jr. and H. M. França. On the connection between the Liouville equation and the Schrödinger equation, 2005. arXiv:quant-ph/0512049.

[8] D. I. Bondar, R. Cabrera, R. R. Lompay, M. Yu. Ivanov, and H. A. Rabitz. Operational dynamic modeling transcending quantum and classical mechanics. *Physical Review Letters*, 109:190403, 2012.

[9] B. O. Koopman. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5):315–318, 1931.

[10] M. J. W. Hall, D.-A. Deckert, and H. M. Wiseman. Quantum phenomena modeled by interactions between many classical worlds. *Physical Review X*, 4:041013, 2014.

[11] A. Kenfack and K. Życzkowski. Negativity of the Wigner function as an indicator of non-classicality. *Journal of Optics B: Quantum and Semiclassical Optics*, 6(10):396–404, 2004.

[12] F. J. Narcowich and R. F. O'Connell. Necessary and sufficient conditions for a phase-space function to be a Wigner distribution. *Physical Review A*, 34(1):1–6, 1986.

[13] J. S. Bell. *Speakable and Unspeakable in Quantum Mechanics*. Cambridge University Press, Cambridge, 1987.

[14] L. D. Landau and E. M. Lifshitz. *Quantum Mechanics: Non-Relativistic Theory*. Pergamon Press, 3rd edition, 1977.

[15] H. Goldstein, C. Poole, and J. Safko. *Classical Mechanics*. Addison-Wesley, 3rd edition, 2002.

[16] Fritz John. *Partial Differential Equations*, volume 1 of *Applied Mathematical Sciences*. Springer-Verlag, New York, 4th edition, 1982.

[17] W. H. Zurek. Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3):715–775, 2003.

[18] E. E. Perepelkin, B. I. Sadovnikov, N. G. Inozemtseva, E. V. Burlakov, and P. V. Afonin. Is the Moyal equation for the Wigner function a quantum analogue of the Liouville equation? *Journal of Statistical Mechanics: Theory and Experiment*, 2023(9):093102, 2023.

[19] G. Strang. On the construction and comparison of difference schemes. *SIAM Journal on Numerical Analysis*, 5(3):506–517, 1968.

[20] M. D. Feit, J. A. Fleck, and A. Steiger. Solution of the Schrödinger equation by a spectral method. *Journal of Computational Physics*, 47(3):412–433, 1982.

[21] M. Nedjalkov, H. Kosina, S. Selberherr, C. Ringhofer, and D. K. Ferry. Unified particle approach to Wigner–Boltzmann transport in small semiconductor devices. *Physical Review B*, 70(11):115319, 2004.

[22] J. M. Sellier. A signed particle formulation of non-relativistic quantum mechanics. *Journal of Computational Physics*, 297:254–265, 2015.

[23] A. C. Hirshfeld and P. Henselder. Deformation quantization in the teaching of quantum mechanics. *American Journal of Physics*, 70(5):537–547, 2002.

[24] C. Wetterich. Quantum entanglement and interference from classical statistics, 2008. arXiv:0809.2671.

[25] A. M. Ozorio de Almeida. Entanglement in phase space, 2009. Lecture Notes in Physics 768, 157–219 (Springer); arXiv:quant-ph/0612029.

[26] A. Blass and Y. Gurevich. Negative probabilities: What they are and what they are for, 2020. arXiv:2009.10552.

[27] J. S. Bell. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1(3):195–200, 1964.

[28] K. Banaszek and K. Wódkiewicz. Nonlocality of the Einstein-Podolsky-Rosen state in the Wigner representation. *Physical Review A*, 58(6):4345–4347, 1998.

[29] J. von Neumann. *Mathematische Grundlagen der Quantenmechanik*. Springer, Berlin, 1932.

[30] H. Nha and H. J. Carmichael. Proposed test of quantum nonlocality for continuous variables. *Physical Review Letters*, 93:020401, 2004.

[31] R. García-Patrón, J. Fiurášek, N. J. Cerf, J. Wenger, R. Tualle-Brouri, and Ph. Grangier. Proposal for a loophole-free Bell test using homodyne detection. *Physical Review Letters*, 93:130409, 2004.

[32] R. P. Feynman. Negative probability. In B. J. Hiley and F. D. Peat, editors, *Quantum Implications: Essays in Honour of David Bohm*, pages 235–248. Routledge & Kegan Paul, London, 1987.

[33] D. Leibfried, D. M. Meekhof, B. E. King, C. Monroe, W. M. Itano, and D. J. Wineland. Experimental determination of the motional quantum state of a trapped atom. *Physical Review Letters*, 77(21):4281–4285, 1996.

[34] R. L. Hudson. When is the Wigner quasi-probability density non-negative? *Reports on Mathematical Physics*, 6(2):249–252, 1974.